

UNIT I | Plants in "Bhartiya Gyan Parampara" (12 Lectures)

1.1 Botany — Definition, Scope and Importance

Definition: Botany (Greek *Botane* = plant) is the branch of Biology dealing with the scientific study of plants — their structure, growth, reproduction, metabolism, diseases and evolutionary relationships. Also called **Plant Biology** or **Phytology**.

Scope — Major Branches:

Branch	What it Studies
Morphology	External form and structure of plants
Anatomy	Internal structure of plant organs
Physiology	Life functions and metabolic processes
Taxonomy	Classification and naming of plants
Ecology	Plants and their environment
Genetics	Heredity and variation in plants
Biochemistry	Chemical processes within plants
Ethnobotany	Plants in human culture and tradition
Economic Botany	Plants of commercial/industrial value

Importance:

- **Food Source:** All food chains begin with plants (rice, wheat, pulses, vegetables).
- **Oxygen:** Photosynthesis by plants maintains atmospheric O₂ balance.
- **Medicine:** >80 % of medicines are plant-derived (Quinine, Morphine, Aspirin).
- **Economy:** Timber, rubber, cotton, jute, spices, perfumes — all from plants.
- **Environment:** Plants prevent soil erosion, regulate rainfall and absorb CO₂.
- **Research:** Mendel used pea plants to discover genetics; biofuels from plants.

1.2 History of Botany in India

Period	Key Developments
Ancient (Vedic) 1500-500 BCE	Plants described in Rigveda & Atharvaveda; classification into trees, shrubs, herbs; Parashar Vrikshayurveda; Charaka & Sushruta Samhita
Medieval (Mughal Era)	Plant gardens (Bagicha) developed; Ain-i-Akbari records plant uses; interest in horticulture
British Period 18th–19th c.	William Roxburgh — "Father of Indian Botany", wrote Flora Indica; J.D. Hooker — Flora of British India (17,000+ spp.)
Modern (Post-1947)	Botanical Survey of India (Kolkata); CSIR/ICAR research; Green Revolution HYV crops; J.C. Bose proved plant sensitivity

1.3 Botanical Concepts in Vedas, Upanishads, Ramayan and Puranas

- Rigveda (~1500 BCE)**
Classifies plants as *Vanaspati* (trees with fruit, no visible flower), *Vriksha* (trees with flower + fruit), *Oshadhi* (medicinal herb) and *Virudh* (climber). The sacred Soma plant is prominently described.
- Atharvaveda**
First systematic text on medicinal plants; 290+ hymns on plants; describes Apamarga, Durva, Pippali; introduces the concept of *Bheshaj* (medicine).
- Upanishads**
Chandogya Upanishad discusses plant consciousness; *Brihadaranyaka Upanishad* states plants are living beings with a soul (Atma).
- Ramayan (Valmiki)**
Mentions 100+ plant species — Ashoka Vatika (*Saraca asoca*), Panchavati forest (Banyan, Peepal, Bel, Amla, Ashoka), Sanjeevani Booti, Jamun, Mango, Coconut.
- Puranas**
Vishnu Purana lists five plant types: Vriksha, Gulma, Lata, Valli, Truna. *Agni* and *Garuda Purana* describe medicinal plants; *Brahma Purana* lists sacred Panchvati trees.

1.4 Contribution of Indian Rishis

Rishi	Key Work	Major Contributions
Parashar	Vrikshayurveda	Oldest plant-science text; classified Dicots (Dwimatruka) & Monocots (Ekamatruka); described plant diseases & treatments
Charak	Charaka Samhita	50 Mahakashaya groups; concept of Dravyaguna (plant properties); different plant parts have different medicinal values
Sushrut	Sushruta Samhita	Classified Vanaspati/Vriksha/Virudh/Oshadhi; described 700+ medicinal plants; used plant fibres as surgical sutures
Surpaal	Vrikshayurveda (Horticulture)	Propagation techniques; treatment of plant diseases with herbal preparations; focus on crop improvement
Kautilya	Arthashastra	Plants of economic & strategic importance; forest conservation as state policy; plant-based medicines & antidotes

1.5 Plants in Traditional Food and Folk Medicine

Spices with Dual Role (Food + Medicine):

Plant	Scientific Name	Medicinal Use
Turmeric	Curcuma longa	Anti-inflammatory, antiseptic, wound healing
Ginger	Zingiber officinale	Digestive aid, nausea, cold & cough
Garlic	Allium sativum	Antibiotic, lowers cholesterol, heart health
Neem	Azadirachta indica	Skin diseases, diabetes, malaria, dental care

Tulsi	Ocimum tenuiflorum	Cough, cold, fever, anti-stress adaptogen
Giloy	Tinospora cordifolia	Immunity booster, dengue fever, diabetes
Amla	Phyllanthus emblica	Richest Vit. C source, hair & eye health
Ashwagandha	Withania somnifera	Stress reducer, improves stamina, insomnia

